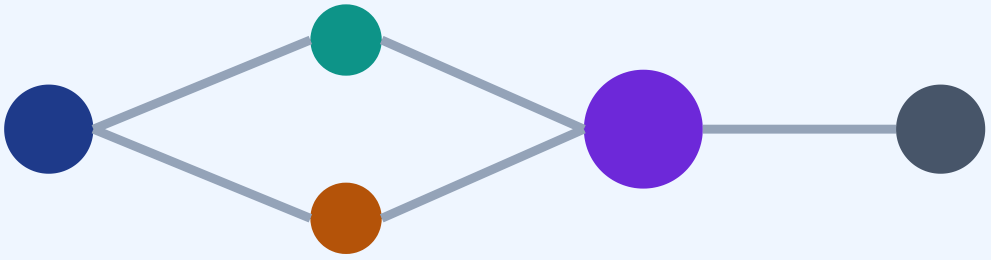
SOFTWARE REQUIREMENTS SPECIFICATION & SYSTEM DESIGN

Hyper-Verified Skill-First Campus Recruitment Architecture (HVS-CRA)



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1. INTRODUCTION & SYSTEM SCOPE

1.1 Purpose of Document

This Software Requirements Specification (SRS) and System Design Document comprehensively defines the architecture, data models, UML designs, and behavioral logic of the **Hyper-Verified Skill-First Campus Recruitment Architecture (HVS-CRA)**. This document serves as the absolute technical blueprint for the engineering teams, quality assurance testers, and regulatory compliance auditors managing the deployment of the AI recruitment engine.

Following the IEEE/ISO 29148 standard, this SRS explicitly outlines both the functional boundaries of the system and the non-functional Service Level Agreements (SLAs) required to process large-scale university placement drives securely and fairly.

1.2 Scope of the HVS-CRA System

The traditional campus hiring landscape experienced a critical breakdown between 2022 and 2026. The proliferation of Generative AI tools allowed graduating candidates to deploy automated bots, submitting hundreds of hyper-optimized, keyword-stuffed resumes across corporate career portals. This "AI vs. AI Arms Race" caused a complete signal collapse; enterprise Applicant Tracking Systems (ATS) became overwhelmed with false-positive matches, while qualified candidates were lost in the algorithmic noise.

The **HVS-CRA** platform replaces the legacy, document-parsing ATS paradigm with a deterministic, verified skill engine. Its scope includes:

- **Cryptographic Ingestion:** Replacing text resumes with Zero-Knowledge (ZK) cryptographic proofs issued by university career centers, validating degrees and coursework without exposing sensitive identity data.
- **Dynamic Assessment Generation:** Executing live, multi-turn coding scenarios that are dynamically synthesized by AI, preventing candidates from using pre-computed LLM answers.
- **Algorithmic Fairness Enforcement:** Processing candidate performance through a Multimodal Counterfactual Debiasing Network (MCDN) to mathematically neutralize biases related to gender, dialect, and university tier.
- **Human-in-the-Loop Orchestration:** Orchestrating real-time offer generation and routing borderline confidence scores to a manual human recruiter queue for final review.

This system operates purely in the B2B SaaS context, bridging Higher Education institutional databases directly with Enterprise Talent Acquisition (TA) pipelines.

1.3 Definitions, Acronyms, and Abbreviations

To ensure precise communication across engineering and regulatory teams, the following terminologies are standardized across all HVS-CRA architecture and UML documentation:

Term / Acronym	Formal Definition & Context in HVS-CRA
ATS	Applicant Tracking System. The legacy enterprise software paradigm being replaced by this system due to its vulnerability to keyword-stuffing.
ZKSV	Zero-Knowledge Skill Verification. A cryptographic protocol (using zk-SNARKs) allowing candidates to prove they possess a degree or skill without revealing their actual transcript or identity.
ALCS	Adaptive Live-Case Synthesis. An AI engine that generates unique technical interview problems in real-time, executing candidate code inside isolated WebAssembly (Wasm) sandboxes.
MCDN	Multimodal Counterfactual Debiasing Network. A specialized neural network layer that transforms raw assessment scores into a debiased latent space, stripping out demographic or socio-economic indicators.
HITL-AG	Human-in-the-Loop Audit Gate. A user interface portal that intercepts algorithmically borderline candidates, forcing a human recruiter to make the final hire/reject decision.
AEDT	Automated Employment Decision Tool. Legal classification (e.g., under NYC Local Law 144) for any software that substantially assists in or makes hiring decisions, mandating strict bias audits.
AIR	Adverse Impact Ratio. An EEOC mathematical standard. Calculated as the selection rate of a protected demographic group divided by the selection rate of the highest-selected group. Must remain ≥ 0.80.
Telemetry Blob	A serialized JSON object containing a candidate's keystroke dynamics, time-to-compile, and inter-key latency, used by the system to detect automated LLM copy-pasting.

2. OVERALL DESCRIPTION & STAKEHOLDERS

2.1 Product Perspective & Legacy Failures

The HVS-CRA system is an autonomous middleware layer. It does not replace the corporate HR Information System (HRIS) (like Workday or SAP), but rather sits in front of it during massive, 48-hour university placement drives. In a legacy environment, 10,000 students might apply for 50 roles, and the ATS would blindly filter based on keywords, resulting in high false positives (candidates who cheated) and false negatives (skilled candidates from lower-tier universities lacking resume polish).

HVS-CRA shifts the paradigm from "filtering text" to "verifying execution." It establishes a direct cryptographic trust bridge with the university, skipping the resume entirely, and routes the verified candidate straight into a live, anti-cheat assessment.

2.2 System Stakeholders & Personas

The system architecture supports four distinct user profiles, each interacting with different bounded contexts of the microservices topology:

Persona / User Class	System Responsibilities & Privileges
Student Candidate	Connects their academic ZK-wallet to the platform. Interacts primarily with the ALCS Sandbox via the web UI to solve dynamic technical challenges. Has no visibility into the scoring algorithms.
Enterprise Recruiter	Configures the multi-criteria decision rubrics prior to a campus drive. Operates the HITL-AG portal to review borderline candidates, analyze execution playbacks, and extend verified job offers.
University Placement Officer	Acts as the root cryptographic authority. Uses the institutional portal to batch-sign student transcripts and skills, enabling students to generate their ZK proofs.
Compliance Auditor	An external or internal legal user with read-only access to the AEDT audit logs. Monitors the Disparate Impact Ratio (DIR) dashboards to ensure the platform operates within EEOC and EU AI Act legal thresholds.

2.3 Operating Environment & Constraints

The HVS-CRA platform is subject to severe technical, regulatory, and physical constraints due to the high-stakes nature of employment automation and the scale of modern placement drives.

1. Strict Latency Constraints:

- Campus drives create massive traffic spikes (e.g., 5,000 students initiating tests simultaneously at 9:00 AM). The ZKSV verification node must cryptographically validate incoming Groth16 proofs in under **150 milliseconds** to prevent connection timeouts.
- The ALCS Wasm sandbox must provision a secure, isolated microVM for a candidate's code execution in under **300 milliseconds**.

2. Regulatory & Data Constraints (AEDT):

- Under the EU AI Act (Annex III), recruitment AI is classified as "High-Risk." The system is constrained to store **zero persistent demographic data** linked to candidate identities. All demographic vectors processed by the MCDN debiasing engine must be held in volatile memory and purged immediately post-computation.
- The system must maintain an immutable, append-only log of every algorithmic decision and every human recruiter override to satisfy NYC Local Law 144 audit requirements.

3. Cryptographic Constraints:

- University nodes must utilize Elliptic Curve Cryptography (specifically the BLS12-381 curve) to sign credentials. The platform rejects any credential signed with outdated algorithms (e.g., RSA-1024 or SHA-1) to protect against future quantum vulnerabilities.

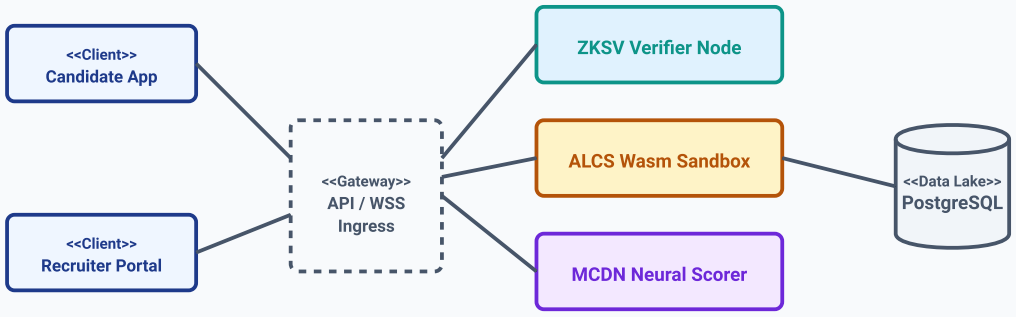
4. Browser & Client Constraints:

- To ensure equitable access for students across varied socio-economic backgrounds, the candidate-facing ALCS assessment UI must be lightweight, rendering fully on browsers supporting ES6 and WebAssembly, without requiring high-end local GPU or CPU resources.

3. SYSTEM ARCHITECTURE & UML COMPONENT DESIGN

The HVS-CRA system utilizes an event-driven microservices architecture deployed on Kubernetes. This topology decouples the heavy computational loads of the AI assessment engines from the synchronous API gateways handling user traffic.

FIGURE 1: UML COMPONENT ARCHITECTURE DIAGRAM



Depicts the ingress flow from external clients, through the API Gateway, routed to domain-specific microservices, and persisted in an AES-256 encrypted database.

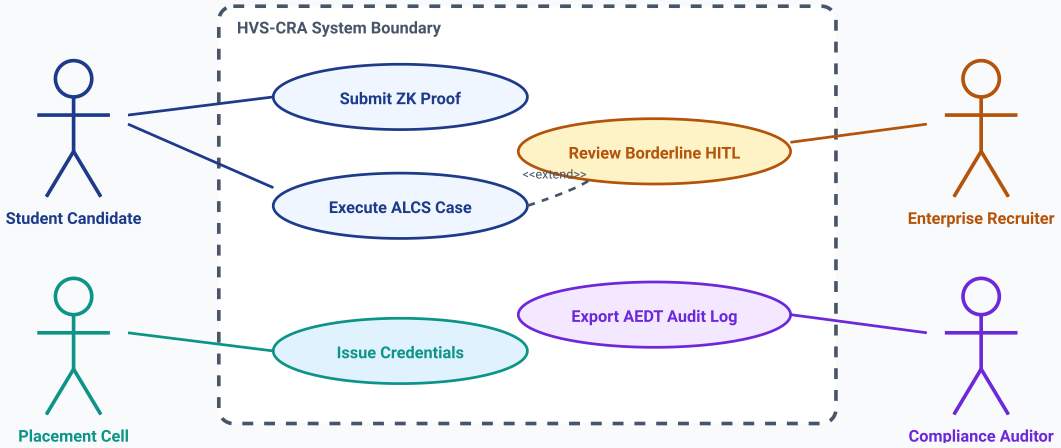
3.1 Component Explanations

The **API Gateway** handles rate-limiting and JWT authentication. It routes synchronous HTTP requests to the **ZKSV Node** for instant credential checks. For live assessments, it establishes a persistent WebSocket connection routing telemetry to the **ALCS Sandbox**. The **MCDN Neural Scorer** operates asynchronously via message queues (e.g., Kafka), processing heavy video/code evaluations in the background before committing the debiased scores to the **PostgreSQL Data Lake**.

4. UML USE CASE MODELING

The UML Use Case Diagram maps the functional interactions between external actors and the system boundaries. It defines what functions the system must perform without dictating how they are implemented.

FIGURE 2: UML USE CASE DIAGRAM



Identifies primary system actors and their permitted functional paths. Extends relationship indicates HITL review is conditionally triggered from ALCS execution.

4.1 Detailed Use Case Specifications

The following tables detail the primary and critical workflows modeled in the UML Use Case diagram.

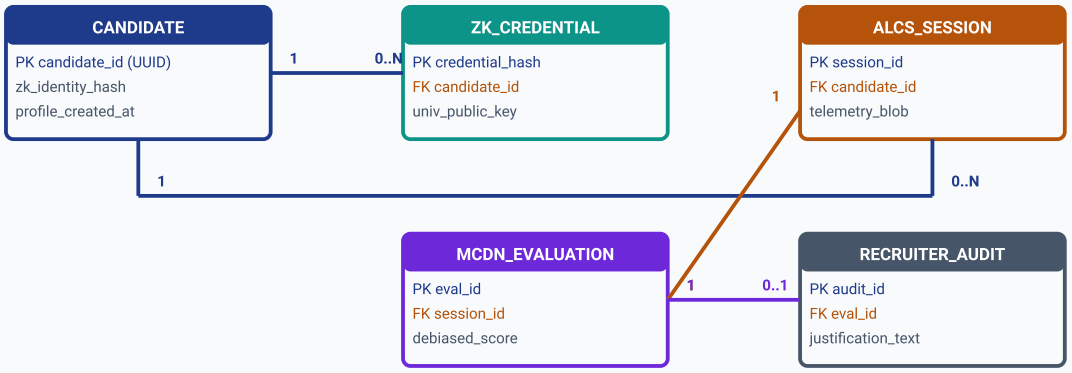
Use Case: UC-01 Submit ZK Proof	
Primary Actor	Student Candidate
Trigger	Candidate logs into the portal and initiates an application for a specific campus drive.
Pre-conditions	Candidate holds a valid, university-signed cryptographic credential wallet.
Main Success Path	1. Candidate selects "Submit Proof". 2. Wallet generates a zk-SNARK payload. 3. ZKSV Node verifies the mathematical proof against the University Public Key. 4. System grants candidate access to the live assessment stage.
Alternative Path	If proof is invalid/tampered, system displays "Verification Failed" and logs IP for anti-bot analysis.

Use Case: UC-04 Review Borderline HITL	
Primary Actor	Enterprise Recruiter
Trigger	MCDN Neural Scorer evaluates an ALCS session and calculates a score within the $[\tau_{low}, \tau_{high}]$ borderline threshold.
Pre-conditions	Candidate has completed the ALCS case. Score is mathematically debiased.
Main Success Path	1. Candidate is placed in the "Requires Review" queue. 2. Recruiter opens the candidate's anonymized telemetry playback. 3. Recruiter reviews code execution logic. 4. Recruiter selects "Extend Offer" or "Reject". 5. Recruiter enters a mandatory justification text.
Post-conditions	Decision and justification are permanently committed to the RECRUITER_AUDIT table for AEDT compliance.

5. DATA DESIGN & ENTITY-RELATIONSHIP DIAGRAM (ERD)

The ERD maps the relational data architecture of the HVS-CRA system. It demonstrates how the system maintains referential integrity while strictly adhering to data-minimization laws (like GDPR and EEOC anonymization mandates) by isolating PII from performance evaluation data.

FIGURE 3: UML ENTITY-RELATIONSHIP DIAGRAM



ERD Detailed Explanation: *The database is highly normalized (3NF). The CANDIDATE table links 1:N to ZK_CREDENTIALS, allowing students to submit multiple skill proofs. It also links 1:N to ALCS_SESSIONs. Importantly, MCDN_EVALUATION links strictly 1:1 to a Session, ensuring every assessment generates exactly one debiased score. The RECRUITER_AUDIT table has a 1:0..1 cardinality with Evaluation, meaning audits only exist if a human recruiter intervened.*

5.2 Detailed Data Dictionary & Integrity Rules

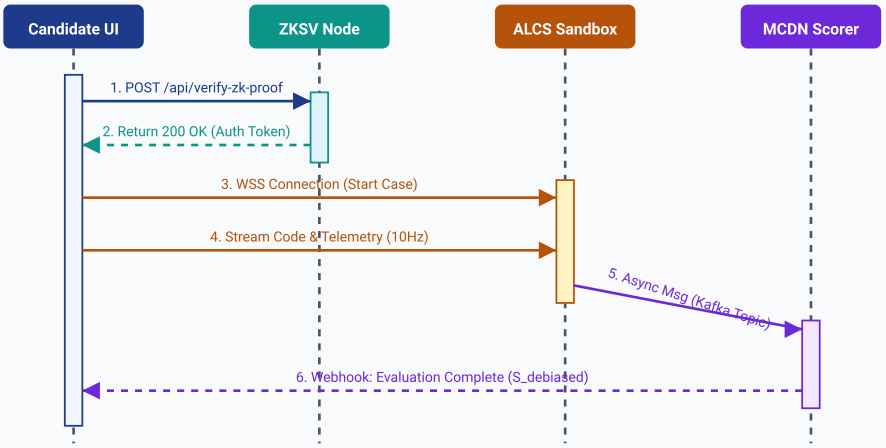
The backend utilizes PostgreSQL 15. The following schema defines the core data types, primary keys (PK), foreign keys (FK), and strict retention policies required for legal AEDT compliance.

Entity / Field	Data Type	Key	Description & Integrity Constraints
Table: CANDIDATE			
candidate_id	UUID	PK	Unique identifier. Auto-generated.
zk_identity_hash	VARCHAR(64)	UNIQUE	SHA-256 hash derived from the candidate's university ZK wallet. Replaces email/name for anonymity.
profile_created_at	TIMESTAMP	-	Standard ISO-8601 timestamp.
Table: ZK_CREDENTIAL			
credential_hash	VARCHAR(64)	PK	Unique hash of the submitted Groth16 mathematical proof.
candidate_id	UUID	FK	References CANDIDATE. ON DELETE CASCADE.
univ_public_key	VARCHAR(128)	-	The ECDSA public key of the university placement cell that signed the credential.
Table: ALCS_SESSION			
session_id	UUID	PK	Unique assessment instance ID.
telemetry_blob	JSONB	-	Contains serialized keystroke dynamics and code compilation metrics. Constraint: Max size 5MB.
Table: MCDN_EVALUATION			
eval_id	UUID	PK	Unique scoring ID.
session_id	UUID	FK	References ALCS_SESSION. ON DELETE CASCADE.
debiased_score	INT	-	CHECK constraint: Must be between 0 and 100.
Table: RECRUITER_AUDIT			
audit_id	UUID	PK	Unique audit log ID.
eval_id	UUID	FK	References MCDN_EVALUATION.
justification_text	TEXT	-	CHECK constraint: `length(justification_text) > 50`. Mandatory for recruiter overrides.

6. BEHAVIORAL DESIGN: UML SEQUENCE DIAGRAM

The UML Sequence Diagram models the chronological order of messages passed between systems to evaluate a single candidate during a campus placement drive. It highlights synchronous API calls and asynchronous processing queues.

FIGURE 4: UML END-TO-END SEQUENCE & MESSAGING MODEL

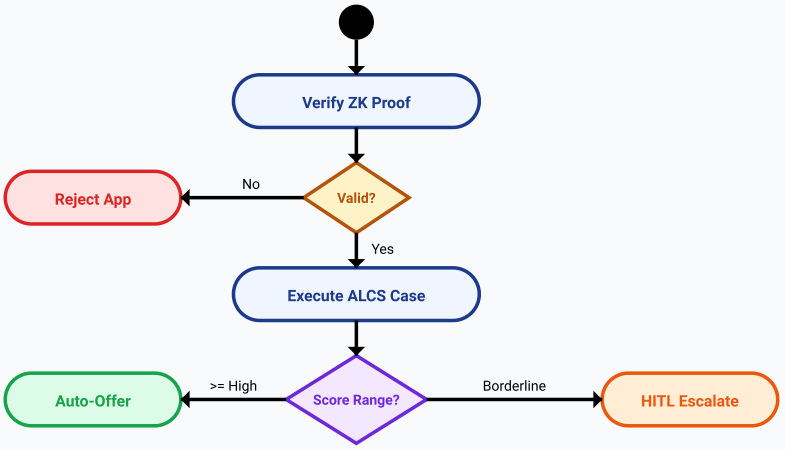


Sequence Explanation: *The workflow strictly separates synchronous UI validation (Steps 1-2) from asynchronous heavy computation. The ZKSV Node validates the mathematical proof instantly. The candidate UI then connects directly via WebSockets (WSS) to the ALCS Sandbox. Once execution completes, the sandbox pushes the payload to a Kafka queue, allowing the MCDN Scorer to process neural transformations entirely offline.*

7. BEHAVIORAL DESIGN: UML ACTIVITY DIAGRAM

The UML Activity (State Flow) Diagram maps the conditional logic and algorithmic decision branching a candidate undergoes during the HVS-CRA placement drive.

FIGURE 5: CANDIDATE EVALUATION STATE FLOW



Activity Explanation: *Illustrates the deterministic flow of a candidate application. The critical routing logic occurs at the final decision node ("Score Range?"). Only candidates whose MCDN scores land precisely in the grey area between deterministic rejection and deterministic offer are escalated to human recruiters, thereby optimizing enterprise TA bandwidth during high-volume drives.*

8. SPECIFIC FUNCTIONAL REQUIREMENTS

This section strictly defines the functional behavior expected from the system's software modules. Requirements are mapped directly to testing IDs for the QA matrix.

8.1 ZKSV & ALCS Subsystems

Req ID	Module	Functional Specification
FR-ZK-001	ZKSV Node	The system shall expose a `/verify` REST endpoint that accepts a JSON payload containing the candidate's Groth16 zk-SNARK string.
FR-ZK-002	ZKSV Node	The system shall validate the cryptographic proof against the globally cached University Public Key Registry.
FR-ZK-003	ZKSV Node	Upon successful validation, the system shall issue a short-lived (60 minute) JSON Web Token (JWT) authorizing the candidate to access the ALCS environment.
FR-AL-001	ALCS Sandbox	The system shall dynamically generate a technical prompt using a Markov Decision Process, ensuring no two candidates in a drive receive the identical prompt.
FR-AL-002	ALCS Sandbox	The UI shall capture keystroke telemetry at 10Hz and serialize it into a `telemetry_blob` JSON object.
FR-AL-003	ALCS Sandbox	The system shall terminate the Wasm sandbox environment forcefully if code compilation memory exceeds 256MB or execution time exceeds 1.5 seconds.
FR-AL-004	ALCS Sandbox	If candidate WSS connection drops, the system shall retain sandbox state for exactly 300 seconds to allow reconnection before wiping volatile memory.

8.2 MCDN & HITL-AG Subsystems

The AI scoring and Recruiter portal modules carry significant legal liability, requiring deterministic logging and access control functional behaviors.

Req ID	Module	Functional Specification
FR-MC-001	MCDN Scorer	The system shall consume Kafka messages containing the `telemetry_blob` and raw execution vectors from the ALCS sandbox.
FR-MC-002	MCDN Scorer	The neural engine shall compute a raw score (S_{raw}) based on code cyclomatic complexity, syntactic correctness, and execution speed.
FR-MC-003	MCDN Scorer	The system MUST pass S_{raw} through the counterfactual debiasing latent space to strip correlated demographic attributes, outputting integer $S_{debiased}$ (0-100).
FR-MC-004	MCDN Scorer	The system shall continuously recalculate the Adverse Impact Ratio (AIR) per drive. If AIR drops below 0.80, it shall trigger an automated P1 engineering alert.
FR-HI-001	HITL Portal	The portal shall fetch and display candidates with scores between configurable $[\tau_{low}, \tau_{high}]$ parameters (default: 65 to 78).
FR-HI-002	HITL Portal	The UI MUST obfuscate the candidate's name, university, and face during the recruiter review to prevent unconscious human bias re-entering the process.
FR-HI-003	HITL Portal	The system shall require the recruiter to select an override reason from a dropdown and input a >50 character justification text before submitting a final decision.

9. NON-FUNCTIONAL REQUIREMENTS (NFR)

System-wide Service Level Agreements (SLAs) ensure the HVS-CRA platform remains operational, secure, and legally compliant during massive volume spikes characteristic of campus hiring drives.

9.1 Performance, Scalability & Availability

- NFR-PERF-01 **Throughput:** The API Gateway (NGINX/Kong) must successfully route 10,000 HTTP Requests Per Second (RPS) without dropping connections during peak morning drive openings.
- NFR-PERF-02 **ZK Latency:** Cryptographic verification of Groth16 proofs must complete with a p95 latency of $\le 200\text{ ms}$.
- NFR-PERF-03 **Auto-Scaling:** Kubernetes Horizontal Pod Autoscaler (HPA) MUST provision new ALCS Wasm sandbox worker nodes within 120 seconds if cluster CPU utilization exceeds 75%.
- NFR-AVAIL-01 **Uptime:** The system architecture must guarantee 99.99% availability during the primary campus recruitment season (September to November).

9.2 Security & Cryptography

- NFR-SEC-01 **Data in Transit:** All external API endpoints and internal gRPC microservice channels must enforce TLS 1.3 encryption.
- NFR-SEC-02 **Data at Rest:** The PostgreSQL persistent data lake must utilize AES-256 block-level encryption managed via AWS Key Management Service (KMS).
- NFR-SEC-03 **JWT Lifetimes:** Authentication JSON Web Tokens (JWT) issued to candidate clients must have a hard expiration limit of 60 minutes and cannot be refreshed.

9.3 Disaster Recovery

- NFR-DR-01 **RTO (Recovery Time Objective):** In the event of a total availability zone failure (e.g., AWS us-east-1 outage), the system must fully failover to a secondary region within 15 minutes.
- NFR-DR-02 **RPO (Recovery Point Objective):** Data loss regarding candidate assessment progress must not exceed 1 minute of telemetry state.

10. EXTERNAL INTERFACE REQUIREMENTS

10.1 Application Programming Interfaces (API)

The system exposes secure RESTful OpenAPI 3.0 endpoints for client communication. The most critical interface is the ZKSV Ingress payload, which securely validates candidate credentials without ingesting PII.

Endpoint: `POST /api/v1/zksv/verify-proof`

Rate Limit: 5 requests / IP / minute (Anti-DDoS measure).

Request Payload (JSON):

```
{
  "candidate_wallet_id": "0x7F2B...a9D",
  "university_domain": "placement.university.edu",
  "zk_proof": {
    "pi_a": ["0x...", "0x..."],
    "pi_b": [["0x...", "0x..."], ["0x...", "0x..."]],
    "pi_c": ["0x...", "0x..."]
  },
  "public_signals": ["1"] // 1 = Degree Verified True
}
```

Response Payload (JSON) - 200 OK:

```
{
  "verification_status": "SUCCESS",
  "access_token": "eyJhbGciOiJIUzI...", // Valid for 60 min
  "sandbox_wss_url": "wss://alcs.hvs-cra.ai/stream/8f2c"
}
```

10.2 User Interface (UI) Accessibility

The Candidate Web UI must comply with **WCAG 2.1 Level AA** standards to ensure equitable access. This includes maintaining a minimum 4.5:1 contrast ratio for text, providing screen-reader compatible ARIA tags for all interactive coding terminal elements, and fully supporting keyboard-only navigation within the ALCS Sandbox.

11. MATHEMATICAL & ALGORITHMIC MODELS

The integrity of the HVS-CRA system relies on strict mathematical formulations governing both cryptographic validation and algorithmic fairness.

11.1 Cryptographic Proof Verification Logic (ZKSV)

The system utilizes Groth16 zk-SNARKs over the BLS12-381 elliptic curve. The ZKSV Node verifies that the candidate possesses a valid degree credential without seeing the transcript. The mathematical pairing check evaluates:

$$e(\pi_A, \pi_B) = e(\alpha, \beta) \cdot e(X, \gamma) \cdot e(\pi_C, \delta)$$

Where `e` represents the optimal Ate pairing function, `(α, β, γ, δ)` are public verification keys from the university's trusted setup, and `X` is the multi-scalar multiplication of the public signals.

11.2 Multimodal Counterfactual Debiasing (MCDN)

To comply with AEDT fairness laws, the raw candidate score vector (X) must be projected into a debiased latent space (Z). The neural network minimizes a joint loss function balancing Task Accuracy (L_{task}) and Counterfactual Fairness (L_{CF}) across a set of protected attributes $A \in \{0, 1\}$ (e.g., gender, university tier):

$$L_{total} = L_{task}(Y, \hat{Y}) + \lambda \cdot E_X [\mid P(\hat{Y}=1 \mid X, A=0) - P(\hat{Y}=1 \mid X, A=1) \mid]$$

Where `λ` is an auto-tuning hyperparameter. The system enforces demographic parity by ensuring that the probability of an "Offer" recommendation ($\hat{Y}=1$) remains mathematically independent of the sensitive attribute `A`.

11.3 Multi-Criteria Decision Analysis (MCDA)

The final placement drive ranking (R) aggregates the verified components. Enterprise recruiters configure the weights (w_i) based on role requirements:

$$R(Candidate_i) = w_1(ZKSV_{bool}) + w_2(S_{debiased}) + w_3(Cadence_{bot_check})$$

12. REGULATORY GOVERNANCE & COMPLIANCE

Because the HVS-CRA generates algorithmic employment decisions, it carries significant legal liability. The software design explicitly integrates mechanisms to satisfy multi-jurisdictional AI and labor laws.

Regulatory Framework	HVS-CRA Architectural Enforcement Strategy
NYC Local Law 144 (AEDT Law)	Requires annual independent bias audits and public posting of impact ratios. **Implementation:** The system contains an automated Disparate Impact Ratio (DIR) calculator that generates standardized audit reports exported directly to the `/compliance` REST endpoint.
EU AI Act (Annex III)	Classifies recruitment and workforce AI systems as "High-Risk." **Implementation:** The architecture satisfies the "Human Oversight" mandate via the HITL-AG portal, preventing the AI from acting as a fully autonomous firing/rejection agent without human verification paths.
EEOC Title VII (Four-Fifths Rule)	Mandates that minority selection rates must not fall below 80% (0.80) of the highest-selected group. **Implementation:** The MCDN network actively monitors the Adverse Impact Ratio (AIR). If AIR drops below 0.82, it automatically triggers engineering alerts to adjust the counterfactual loss weight (λ).
GDPR / CCPA / CPRA	Candidates hold the "Right to be Forgotten" and rights against automated profiling. **Implementation:** The ZKSV design inherently supports this by never storing raw PII, only cryptographic hashes. Deleting the single `candidate_id` cascades and anonymizes all performance telemetry instantly.

13. TRACEABILITY MATRIX & OFFICIAL SIGN-OFF

13.1 Requirements Traceability Matrix (RTM) Partial

Ensures that every defined requirement is mapped to a specific system module and covered by the automated QA testing suite prior to production deployment.

Req ID	Subsystem Module	Target API / Component	Automated Test Suite ID
FR-ZK-001	ZKSV Node	`POST /verify-proof`	`test_zksv_groth16_valid`
FR-AL-003	ALCS Sandbox	Wasm Executor	`test_alcs_oom_kill_256mb`
FR-MC-003	MCDN Scorer	Python PyTorch Model	`test_mcdn_air_threshold`
FR-HI-003	HITL-AG Portal	Recruiter React UI	`test_ui_mandatory_justification`
NFR-PERF-01	API Gateway	Kong Ingress	`load_test_10k_rps_locust`

13.2 Engineering & Architecture Sign-Off

This document constitutes the final, frozen Software Requirements Specification (V4.0) for the HVS-CRA platform. All subsequent sprint planning, microservice scaffolding, and QA pipeline generation must strictly adhere to these specifications.

Lead Systems Architect
Team Code Avengers

AEDT Compliance Auditor
Legal Governance Committee